

Fundamentals of Photovoltaics (EE336)

Lecture 4 – Solar Cell Operation

Solar Cell Structure

A solar cell is an electronic device which directly converts sunlight into electricity. Light shining on the solar cell produces both a current and a voltage to generate electric power. This process requires firstly, a material in which the absorption of light raises an electron to a higher energy state, and secondly, the movement of this higher energy electron from the solar cell into an external circuit. The electron then dissipates its energy in the external circuit and returns to the solar cell. A variety of materials and processes can potentially satisfy the requirements for photovoltaic energy conversion, but in practice nearly all photovoltaic energy conversion uses semiconductor materials in the form of a p - n junction.

The basic steps in the operation of a solar cell are:

- the generation of photo-generated carriers;
- the collection of the photo-generated carries to generate a current;
- the generation of a large voltage across the solar cell; and
- the dissipation of power in the load and in parasitic resistances.

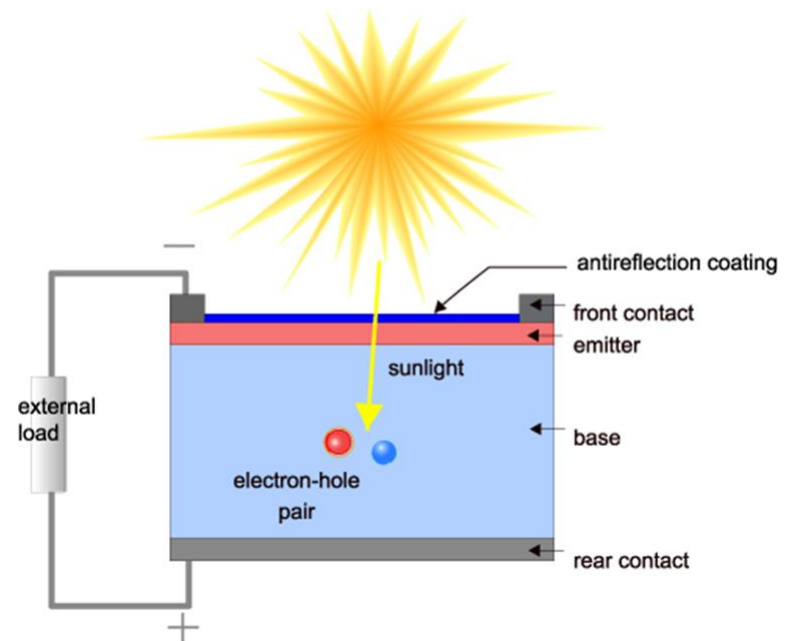


Photo-generated current

- The generation of current in a solar cell, known as the “photo-generated current”, involves two key processes. The first process is the absorption of incident photons to create electron-hole pairs. Electron-hole pairs will be generated in the solar cell provided that the incident photon has an energy greater than that of the band gap. However, electrons (in the p -type material), and holes (in the n -type material) are **meta-stable** and will only exist, on average, for a length of time equal to the minority carrier lifetime before they recombine. If the carrier recombines, then the light-generated electron-hole pair is lost and no current or power can be generated.
- A second process, the collection of these carriers by the p - n junction, prevents this recombination by using a p - n junction to spatially separate the electron and the hole. The carriers are separated by the action of the electric field existing at the p - n junction. If the light-generated minority carrier reaches the p - n junction, it is swept across the junction by the electric field at the junction, where it is now a majority carrier. If the emitter and base of the solar cell are connected together (i.e., if the solar cell is short-circuited), the photo-generated carriers flow through the external circuit.

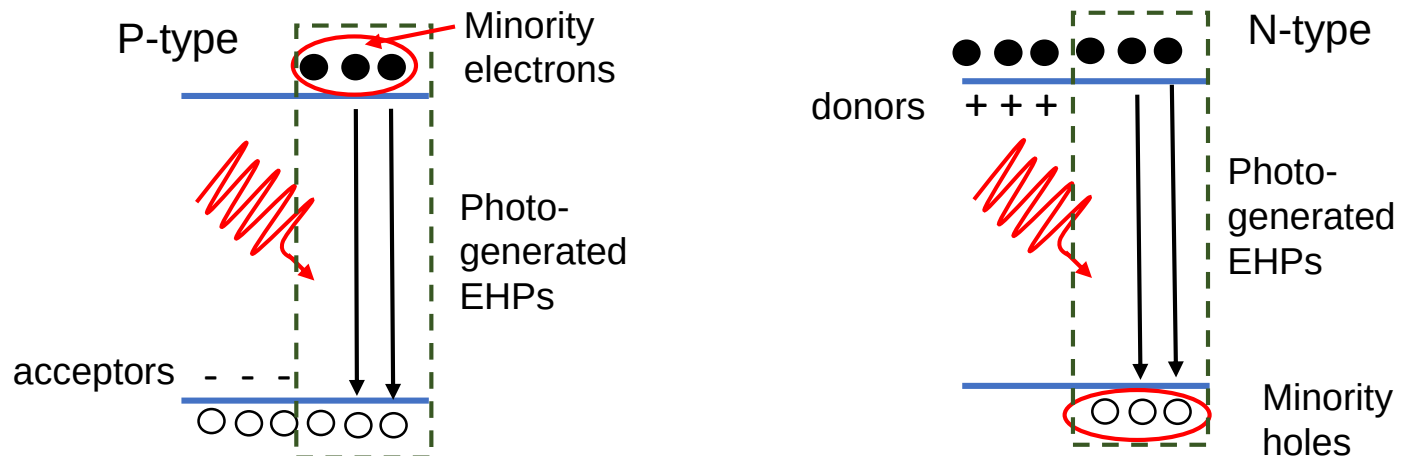
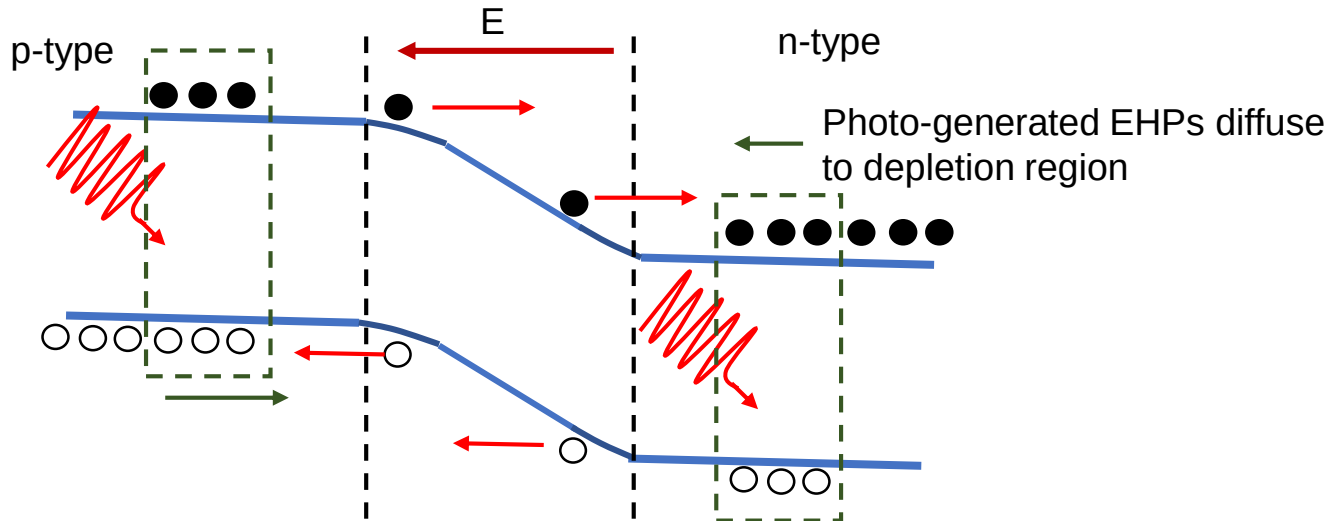


Photo-generated current

P-N Junction

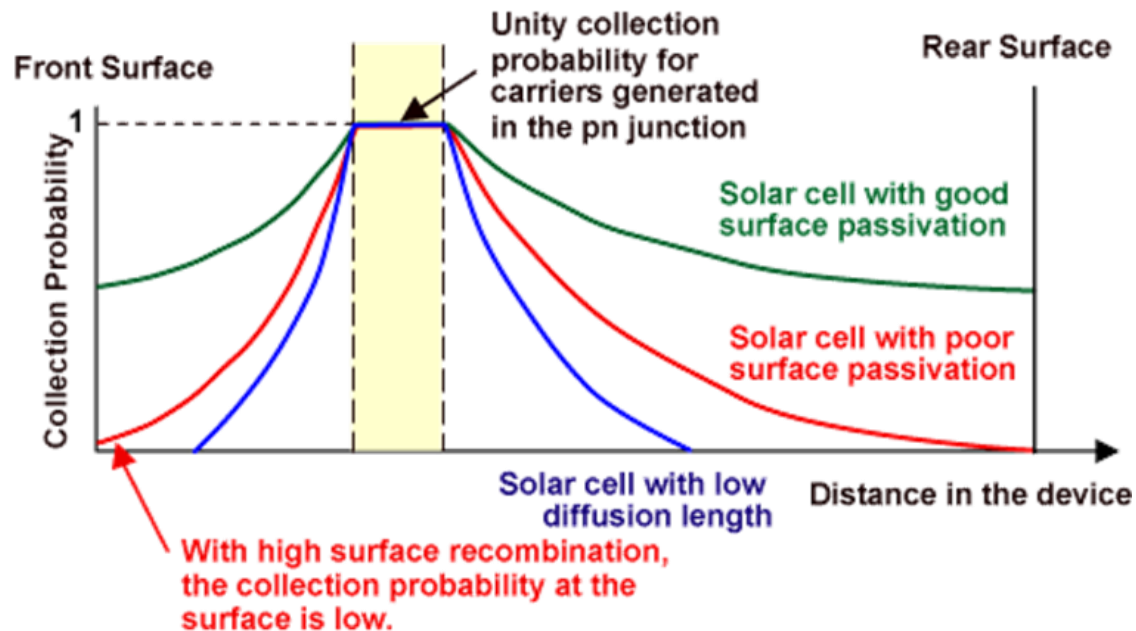


1. The light is absorbed in P and N region and EHPs are generated.
2. EHPs generated within the diffusion length are able to reach the depletion region.
3. In the space-charge region, the electrons are swept to n-side and holes to p-side, becoming “free” majority carriers.

Collection Probability

- The "collection probability" describes the probability that a carrier generated by light absorption in a certain region of the device will be collected by the p - n junction and therefore contribute to the light-generated current, but probability depends on the distance that a light-generated carrier must travel compared to the diffusion length. Collection probability also depends on the surface properties of the device.
- The collection probability of carriers generated in the depletion region is unity as the electron-hole pair are quickly swept apart by the electric field and are collected.
- Away from the junction, the collection probability drops. If the carrier is generated more than a diffusion length away from the junction, then the collection probability of this carrier is quite low.
- Similarly, if the carrier is generated closer to a region such as a surface with higher recombination than the junction, then the carrier will recombine.

The impact of surface passivation and diffusion length on collection probability is illustrated below.



The collection probability in conjunction with the generation rate in the solar cell determine the light-generated current from the solar cell. The light-generated current is the integration over the entire device thickness of the generation rate at a particular point in the device, multiplied by the collection probability at that point. The equation for the light-generated current density (J_L), with an arbitrary generation rate ($G(x)$) and collection probability ($CP(x)$), is shown below, as is the generation rate in silicon due to the AM1.5 solar spectrum:

$$J_L = q \int_0^W G(x) CP(x) dx = q \int_0^W \left[\int \alpha(\lambda) H_0 \exp(-\alpha(\lambda)x) d\lambda \right] CP(x) dx \quad (4.1)$$

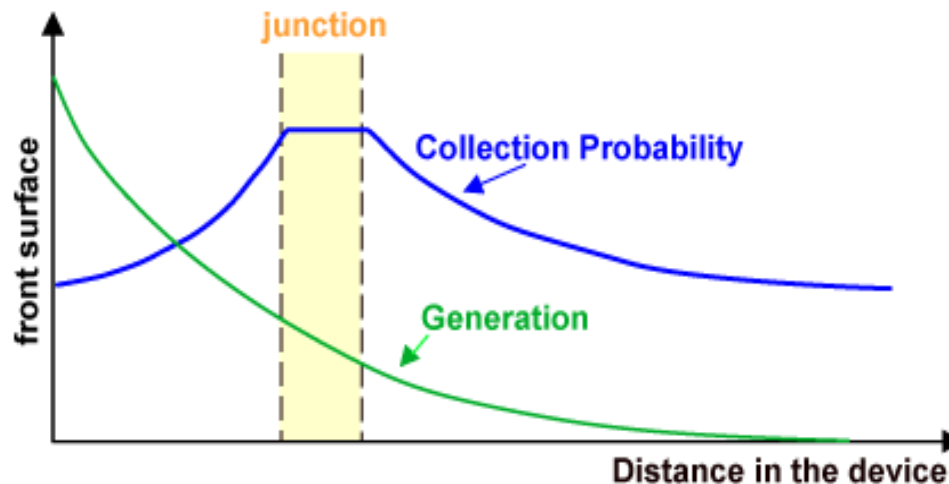
where:

q is the electronic charge;

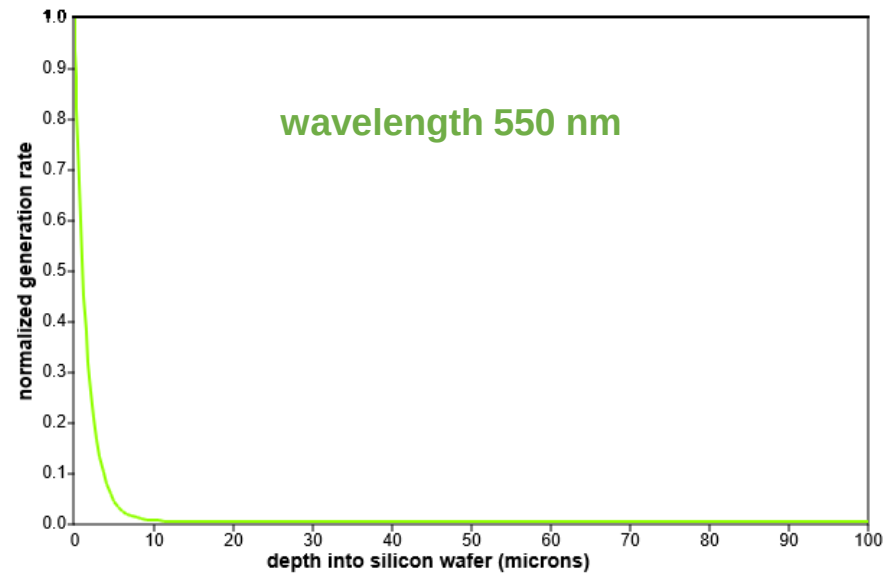
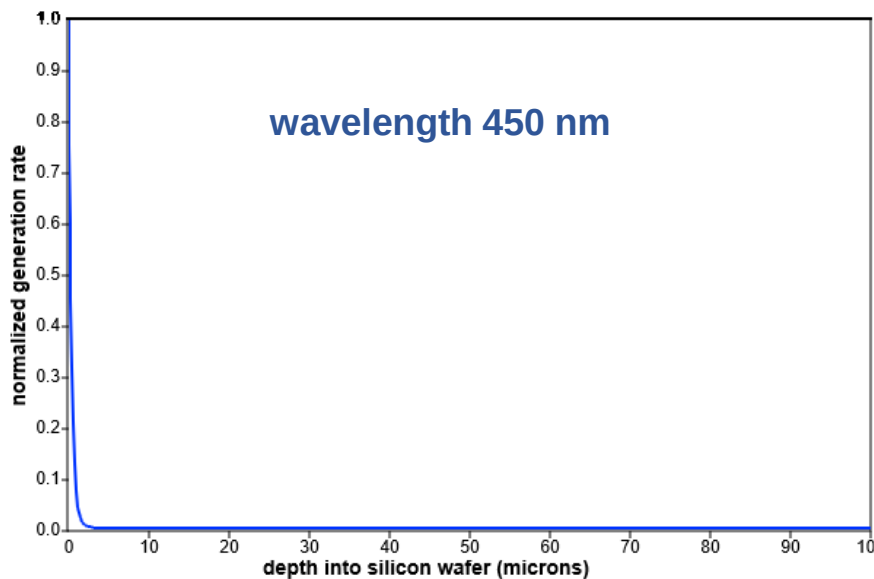
W is the thickness of the device;

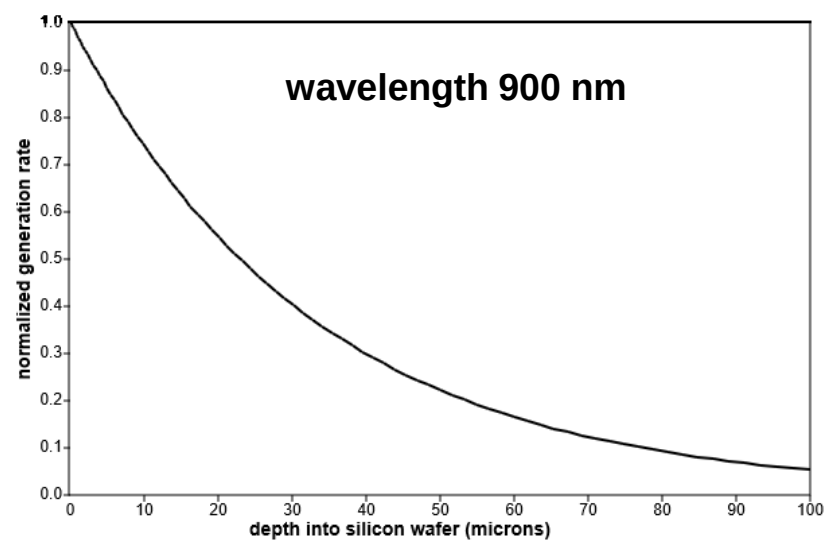
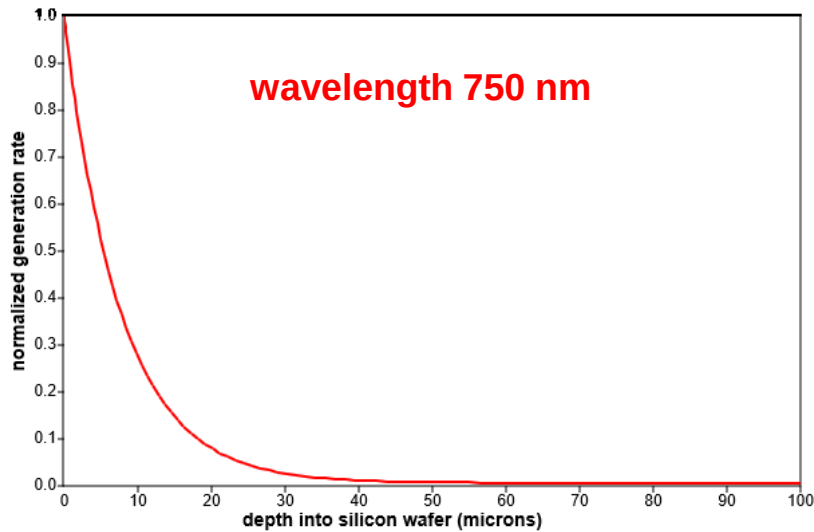
$\alpha(\lambda)$ is the absorption coefficient;

H_0 is the number of photons at each wavelength.



A non-uniform collection probability will cause a spectral dependence in the photo-generated current. For example, at the surfaces, the collection probability is lower than in the bulk. Comparing the generation rates for blue (450 nm), green (550 nm), red (750 nm) and near infrared light (900 nm) below, blue light is nearly completely absorbed in the first few tenths of a micron in silicon. Therefore, if the collection probability at the front surface is low, any blue light in the solar spectrum does not contribute to the light-generated current.





The creation of electron-hole pairs for different wavelengths of light in silicon. Blue light of $0.45\ \mu\text{m}$ has a high absorption coefficient of $10^5\ \text{cm}^{-1}$ and is therefore absorbed very close to the front surface. Red light at $0.8\ \mu\text{m}$ and an absorption coefficient of $10^3\ \text{cm}^{-1}$ is absorbed deeper into the cell. Infrared light at $1.1\ \mu\text{m}$ with an absorption coefficient of $10^3\ \text{cm}^{-1}$ is barely absorbed since it is close to the band gap of silicon.

Quantum Efficiency

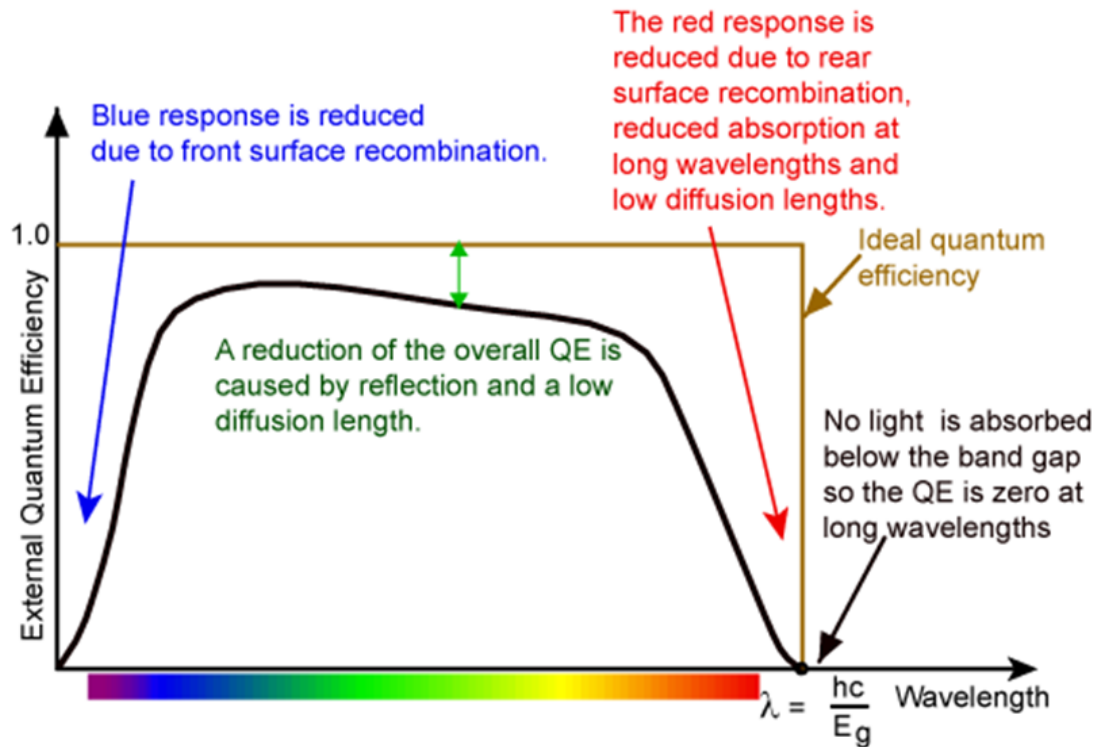
- The “external quantum efficiency” (EQE) is the ratio of the number of carriers collected by the solar cell to the number of photons of a given energy incident on the solar cell. The quantum efficiency may be given either as a function of wavelength or as energy.

$$\eta_{ext}(\lambda) = \frac{N_{electrons}(\lambda)}{N_{photons}(\lambda)} = \frac{I(\lambda)/q}{P_{in}(\lambda)/(hc/\lambda)} = \frac{I(\lambda)hc}{\lambda P_{in}(\lambda)q} = \frac{1240 \cdot I(\lambda)}{\lambda(nm)P_{in}(\lambda)} \quad (4.2)$$

- where q is the charge of electron, c is speed of light, h is Plank’s constant, $I(\lambda)$ is the current generated by the cell at wavelength λ and $P_{in}(\lambda)$ is the incoming power at wavelength λ . EQE is also known as **incident photon to current conversion efficiency (IPCE)**. Given the spectrum of the incident light $S(\lambda)$ and , the photocurrent can be calculated using:

$$J_{ph} = \int \frac{q\lambda}{hc} \eta_{ext}(\lambda) S(\lambda) d\lambda \quad (4.3)$$

- If all photons of a certain wavelength are absorbed and the resulting minority carriers are collected, then the quantum efficiency at that particular wavelength is unity. The quantum efficiency for photons with energy below the band gap is zero. A quantum efficiency curve for an ideal solar cell is shown below.



The quantum efficiency of a silicon solar cell. Quantum efficiency is usually not measured much below 350 nm as the power from the AM1.5 contained in such low wavelengths is low.

While quantum efficiency ideally has the square shape shown above, the quantum efficiency for most solar cells is reduced due to recombination effects.

The same mechanisms which affect the collection probability also affect the quantum efficiency.

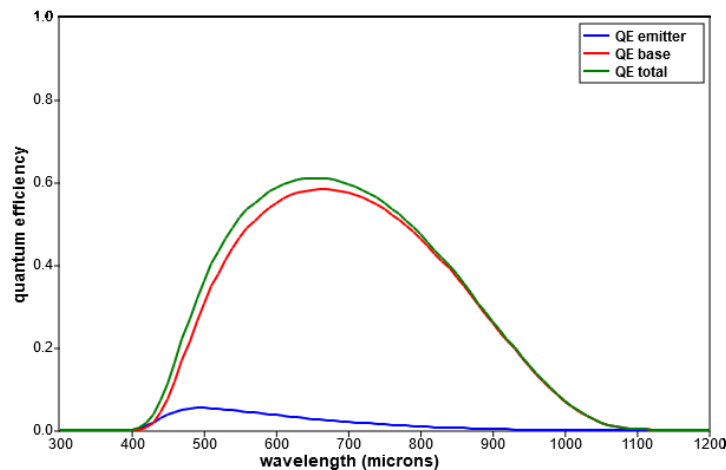
For example, front surface passivation affects carriers generated near the surface, and since blue light is absorbed very close to the surface, high front surface recombination will affect the "blue" portion of the quantum efficiency.

Similarly, green light is absorbed in the bulk of a solar cell and a low diffusion length will affect the collection probability from the solar cell bulk and reduce the quantum efficiency in the green portion of the spectrum.

The quantum efficiency can be viewed as the collection probability due the generation profile of a single wavelength, integrated over the device thickness and normalized to the incident number of photons.

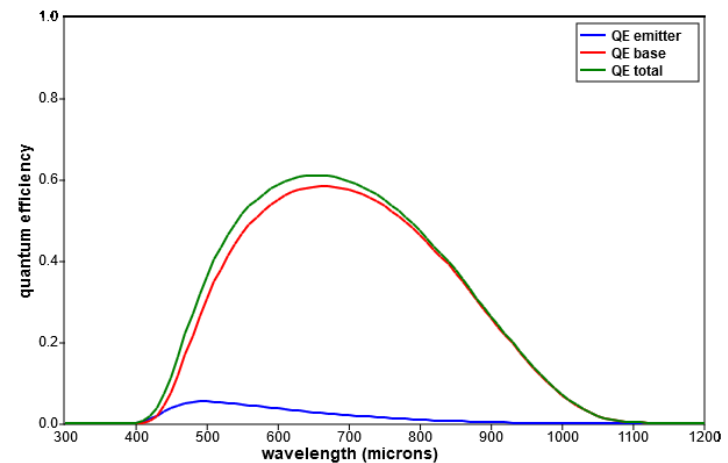
The figure below shows the effect on surface recombination and diffusion length on the internal quantum efficiency of a solar cell. The emitter thickness is 1 μm , the base thickness is 300 μm , the emitter diffusivity is 4 cm^2s^{-1} and the base diffusivity is 27 cm^2s^{-1} . For base diffusion lengths greater than the device thickness of 300 μm the rear surface recombination velocity has a large effect on QE. For low diffusion lengths recombination at the rear surface has no effect.

Go to www.pveducation.org and observe the effect of diffusion length and surface recombination on the QE of a silicon solar cell.



Base diffusion length: 12 microns

Rear surface recombination velocity: 8.5
cm/s



Base diffusion length: 12 microns

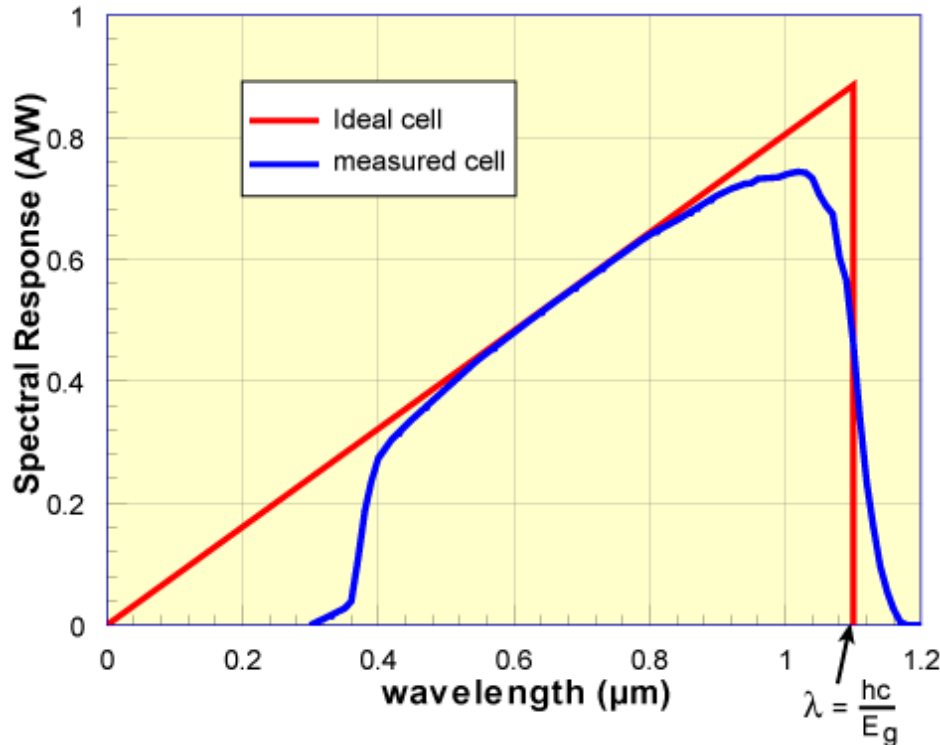
Rear surface recombination velocity: 5370.3
cm/s

- In some cases, it may be preferable to express an efficiency of a solar cell without considering its absorption efficiency (in other word, without considering the effect of optical losses such as transmission and reflection). Internal quantum efficiency (IQE or $\eta_{\text{IQE}}(\lambda)$) is more appropriate way to represent the efficiency in such case. IQE is the ratio of the number of collected charges at the electrode to the number of absorbed photons. It is related with η_{EQE} by absorption efficiency.

$$\eta_{\text{EQE}} = \eta_{\text{A}} \eta_{\text{IQE}} \quad (4.4)$$

Spectral Response

The spectral response is conceptually similar to the quantum efficiency. The quantum efficiency gives the number of electrons output by the solar cell compared to the number of photons incident on the device, while the spectral response is the ratio of the current generated by the solar cell to the **power** incident on the solar cell. A spectral response curve is shown below.



The spectral response of a silicon solar cell under glass. At short wavelengths below 400 nm the glass absorbs most of the light and the cell response is very low. At intermediate wavelengths the cell approaches the ideal. At long wavelengths the response fall back to zero. Silicon is an indirect band gap semiconductor so there is not a sharp cut off at the wavelength corresponding to the band gap ($E_g = 1.12$ eV).

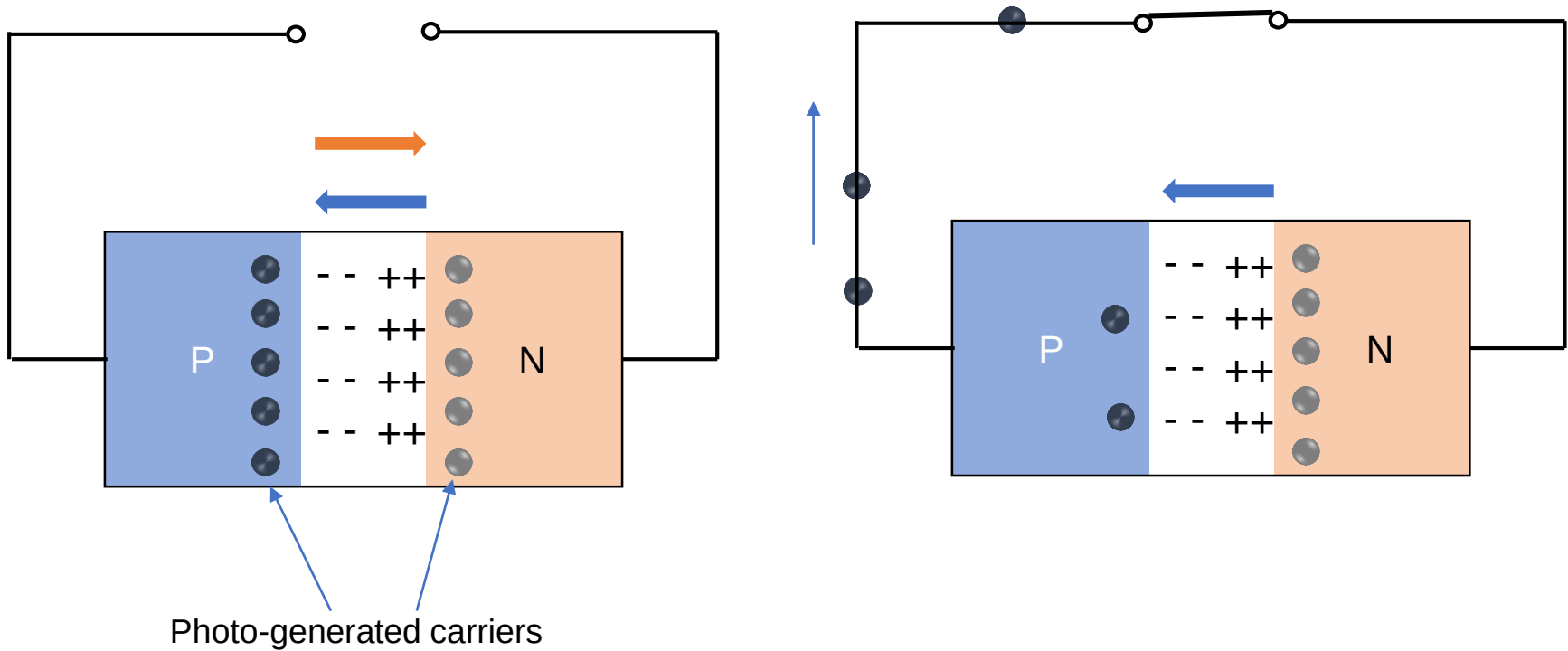
- The ideal spectral response is limited at long wavelengths by the inability of the semiconductor to absorb photons with energies below the band gap. This limit is the same as that encountered in quantum efficiency curves. However, unlike the square shape of QE curves, the spectral response decreases at small photon wavelengths. At these wavelengths, each photon has a large energy, and hence the ratio of photons to power is reduced. Any energy above the band gap energy is not utilized by the solar cell and instead goes to heating the solar cell. The inability to fully utilize the incident energy at high energies, and the inability to absorb low energies of light represents a significant power loss in solar cells consisting of a single *p-n* junction.
- Spectral response is important since it is the spectral response that is measured from a solar cell, and from this the quantum efficiency is calculated. The quantum efficiency can be determined from the spectral response by replacing the power of the light at a particular wavelength with the photon flux for that wavelength. This gives:

$$SR = \frac{q\lambda}{hc} QE \quad (4.5)$$

$$SR (A/W) = \frac{QE \cdot \lambda (nm)}{1240} \quad (4.6)$$

The Photovoltaic Effect

- The collection of light-generated carriers does not by itself give rise to power generation. In order to generate power, a voltage must be generated as well as a current. Voltage is generated in a solar cell by a process known as the "photovoltaic effect". The collection of photo-generated carriers by the p - n junction causes a movement of electrons to the n -type side and holes to the p -type side of the junction. Under short circuit conditions, there is no build up of charge, as the carriers exit the device as photo-generated current.
- However, if the photo-generated carriers are prevented from leaving the solar cell, then the collection of photo-generated carriers causes an increase in the number of electrons on the n -type side of the p - n junction and a similar increase in holes in the p -type material. This separation of charge creates an electric field at the junction which is in opposition to that already existing at the junction, thereby reducing the net electric field. Since the electric field represents a barrier to the flow of the forward bias diffusion current, the reduction of the electric field increases the diffusion current. A new equilibrium is reached in which a voltage exists across the p - n junction. The current from the solar cell is the difference between I_L and the forward bias current. Under open circuit conditions, the forward bias of the junction increases to a point where the photo-generated current is exactly balanced by the forward bias diffusion current, and the net current is zero. The voltage required to cause these two currents to balance is called the "open-circuit voltage".



← Electric field due to ionized donors/acceptors

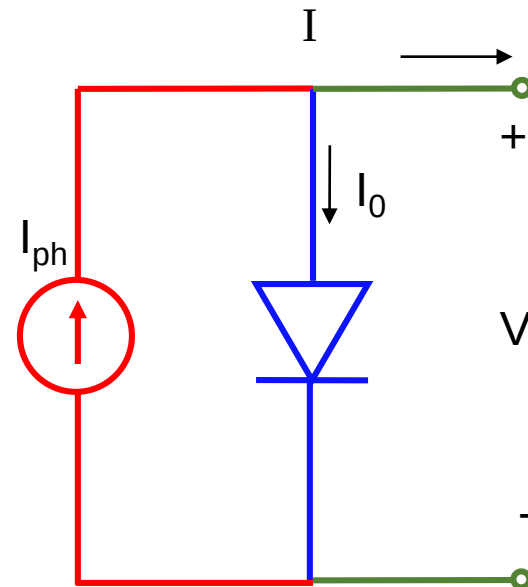
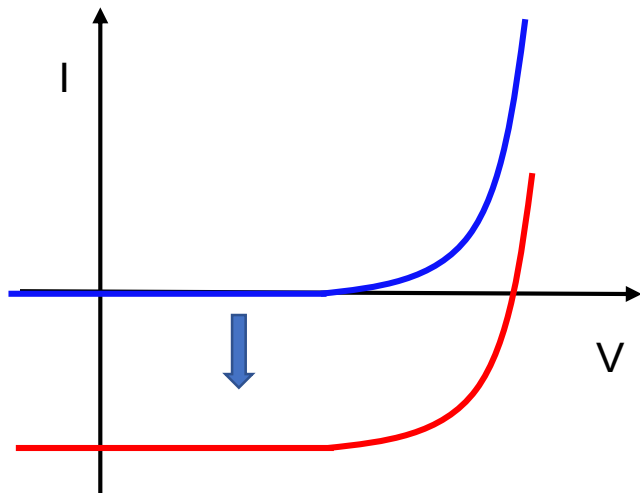
→ Electric field due to photo-generated carriers

Current-voltage (I - V) characteristic

- The I - V curve of a solar cell is the superposition of the I - V curve of the solar cell diode in the dark with the photo-generated current. The light has the effect of shifting the I - V curve down into the fourth quadrant where power can be extracted from the diode. Illuminating a cell adds to the normal "dark" currents in the diode so that the diode equation becomes:

$$I = I_0 \left[e^{\frac{qV}{nkT}} - 1 \right] - I_{ph} \quad (4.7)$$

- where I_{ph} = photo-generated current.

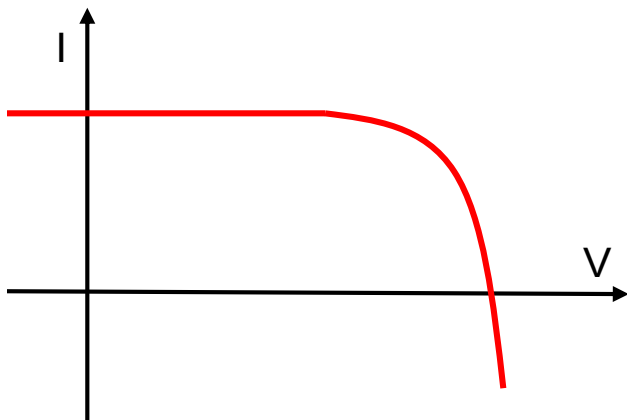


- Since a solar cell is generating power, sometimes its current is preferred to be shown in the first quadrant and invert the current axis.
- The equation for I-V curve in the first quadrant is

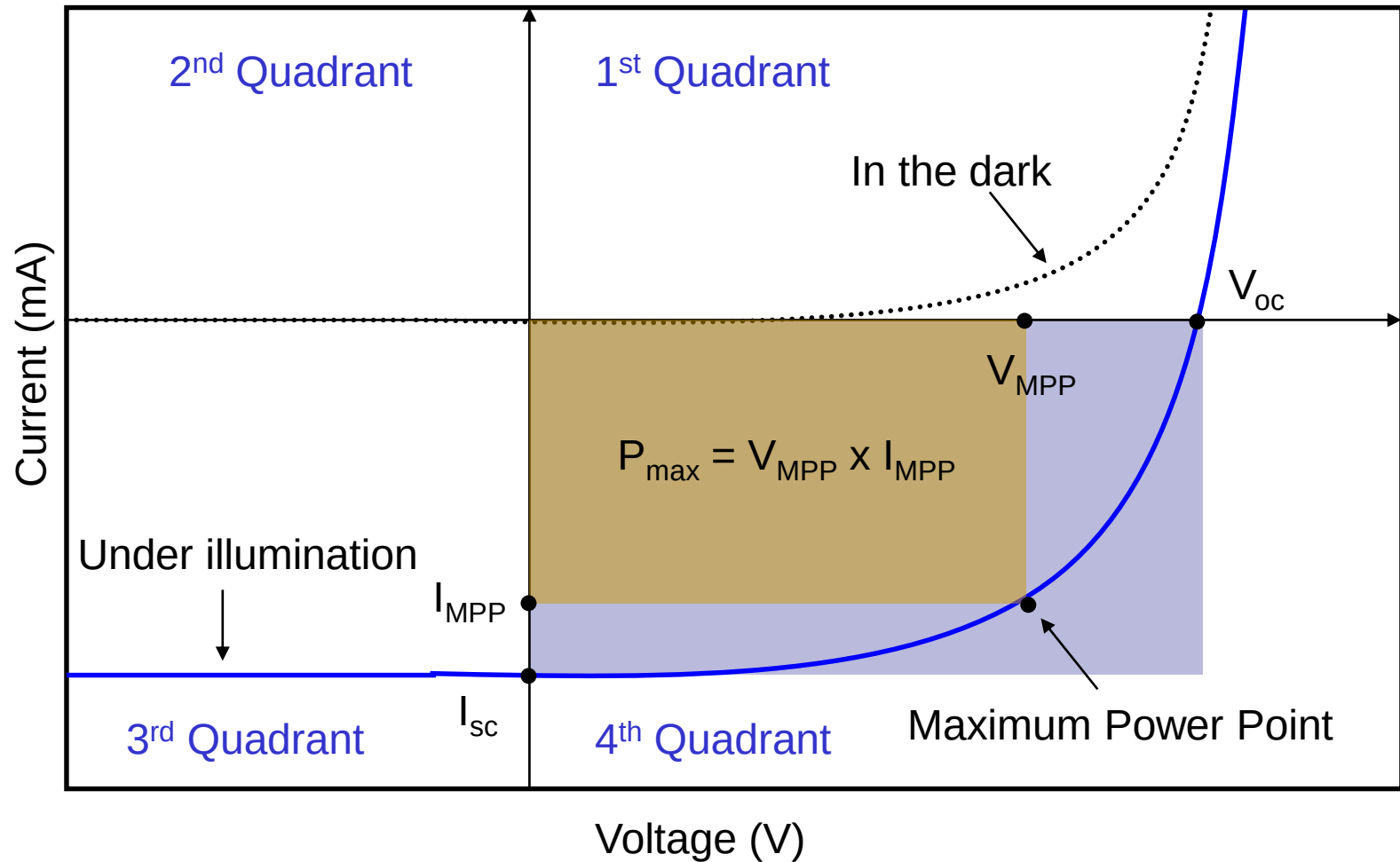
$$I = I_{ph} - I_0 \left[e^{\frac{qV}{nkT}} - 1 \right] \quad (4.8)$$

- The -1 term in the above equation can usually be neglected. The exponential term is usually $\gg 1$ except for voltages below 100 mV. Further, at low voltages the light generated current I_{ph} dominates the I_0 (...) term so the -1 term is not needed under illumination.

$$I = I_{ph} - I_0 \left[e^{\frac{qV}{nkT}} \right] \quad (4.9)$$

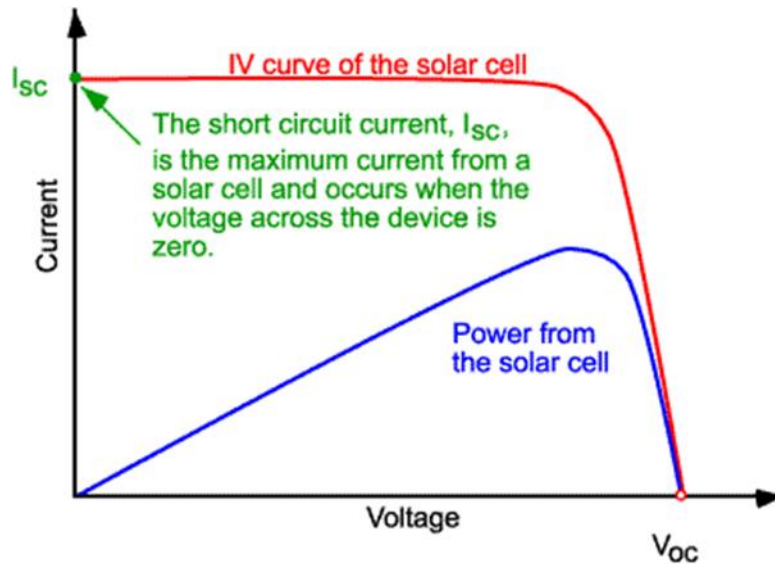


Solar Cell Parameters



Short-circuit current (I_{SC})

- The short-circuit current is the current through the solar cell when the voltage across the solar cell is zero (i.e., when the solar cell is short circuited). Usually written as I_{SC} , the short-circuit current is shown on the I-V curve below.



The short-circuit current is due to the generation and collection of light-generated carriers. For an ideal solar cell at most moderate resistive loss mechanisms, the short-circuit current and the photo-generated current are identical. Therefore, the short-circuit current is the largest current which may be drawn from the solar cell.

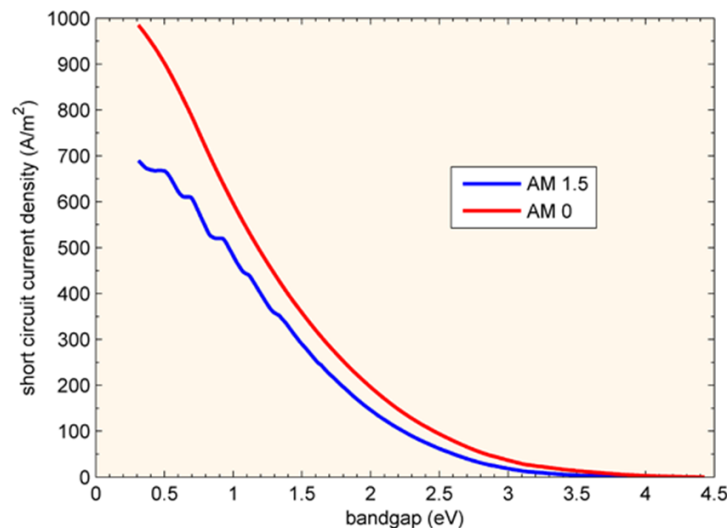
The short-circuit current depends on a number of factors which are described below:

- **the area of the solar cell.** To remove the dependence of the solar cell area, it is more common to list the short-circuit current **density** (J_{sc} in mA/cm²) rather than the short-circuit current;
- **the number of photons** (i.e., the power of the incident light source). I_{sc} from a solar cell is directly dependant on the light intensity
- **the spectrum of the incident light.** For most solar cell measurement, the spectrum is standardised to the AM1.5 spectrum;
- **the optical properties** (absorption and reflection) of the solar cell;
- **the collection probability** of the solar cell, which depends chiefly on the surface passivation and the minority carrier lifetime in the base.

- When comparing solar cells of the same material type, the most critical material parameter is the diffusion length and surface passivation. In a cell with perfectly passivated surface and uniform generation, the equation for the short-circuit current can be approximated as:

$$J_{SC} = qG(L_n + L_p) \quad (4.10)$$

- where G is the generation rate, and L_n and L_p are the electron and hole diffusion lengths respectively. Although this equation makes several assumptions which are not true for the conditions encountered in most solar cells, the above equation nevertheless indicates that the short-circuit current depends strongly on the generation rate and the diffusion length.
- Silicon solar cells under an AM1.5 spectrum have a maximum possible current of 46 mA/cm². Laboratory devices have measured short-circuit currents of over 42 mA/cm², and commercial solar cell have short-circuit currents between about 28 mA/cm² and 35 mA/cm².



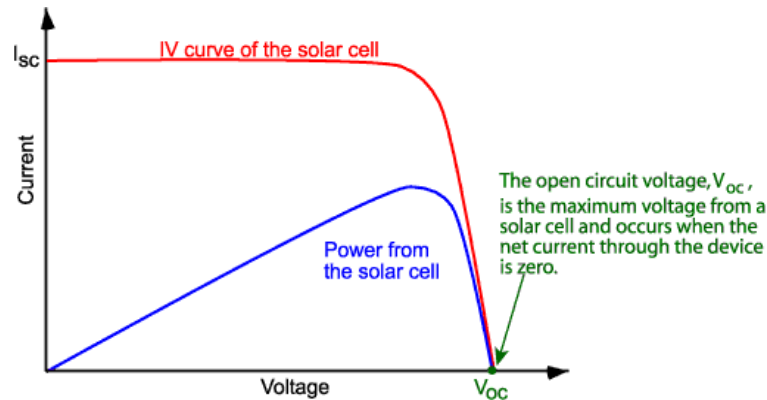
In an ideal device every photon above the bandgap gives one charge carrier in the external circuit so the highest current is for the lowest bandgap.

Photo-generated Current and Short Circuit Current (I_{ph} or I_{sc} ?)

- I_{ph} is the photo-generated current inside the solar cell and is the correct term to use in the solar cell equation. At short circuit conditions the externally measured current is I_{sc} . Since I_{sc} is usually equal to I_{ph} , the two are used interchangeably and for simplicity and the solar cell equation is written with I_{sc} in place of I_{ph} . In the case of very high series resistance ($> 10 \Omega\text{cm}^2$) I_{sc} is less than I_{ph} and writing the solar cell equation with I_{sc} is incorrect.
- Another assumption is that the illumination current I_{ph} is solely dependent on the incoming light and is independent of voltage across the cell. However, I_{ph} varies with voltage in the case of drift-field solar cells and where carrier lifetime is a function of injection level such as defected multi-crystalline materials.

Open-circuit Voltage (V_{oc})

- The open-circuit voltage, V_{oc} , is the maximum voltage available from a solar cell, and this occurs at zero current. The open-circuit voltage corresponds to the amount of forward bias on the solar cell due to the bias of the solar cell junction with the photo-generated current. The open-circuit voltage is shown on the I-V curve below.



- In this condition, there is no current flows between the two terminals of the device, i.e. $I = 0$ and $V = V_{oc}$. From the equation (4.7)

$$I_{ph} - I_0 \left[e^{\frac{qV_{oc}}{nkT}} - 1 \right] = 0 \quad (4.11)$$

$$V_{oc} = \frac{nkT}{q} \ln \left(\frac{I_{ph}}{I_0} + 1 \right) \approx \frac{nkT}{q} \ln \left(\frac{I_{ph}}{I_0} \right) \quad (4.12)$$

- The equation (4.12) shows that V_{oc} depends on the saturation current of the solar cell and the photo-generated current. While I_{ph} typically has a small variation, the key effect is the saturation current, since this may vary by orders of magnitude. The saturation current, I_0 depends on recombination in the solar cell. Open-circuit voltage is then a measure of the amount of recombination in the device. Silicon solar cells on high quality single crystalline material have open-circuit voltages of up to 730 mV under one sun and AM1.5 conditions, while commercial devices on multi-crystalline silicon typically have open-circuit voltages around 600 mV.
- The V_{oc} can also be determined from the carrier concentration:

$$V_{oc} = \frac{kT}{q} \ln \left[\frac{(N_A + \Delta n) \Delta n}{n_i^2} \right] \quad (4.13)$$

where kT/q is the thermal voltage, N_A is the doping concentration, Δn is the excess carrier concentration and n_i is the intrinsic carrier concentration.

V_{oc} as a Function of Bandgap, E_G

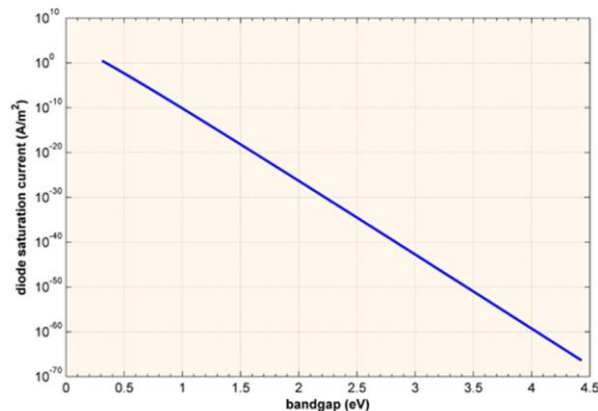
- Where the short-circuit current (I_{SC}) decreases with increasing bandgap, the open-circuit voltage increases as the band gap increases. In an ideal device the V_{OC} is limited by radiative recombination and the analysis uses the principle of detailed balance to determine the minimum possible value for J_0 .
- The minimum value of the diode saturation current is given by

$$J_0 = \frac{q}{k} \frac{15\sigma}{\pi^4} T^3 \int_u^\infty \frac{x^2}{e^x - 1} dx \quad (4.14)$$

where q is the electronic charge, σ is the Stefan–Boltzman constant, k is Boltzmann constant, T is the temperature and

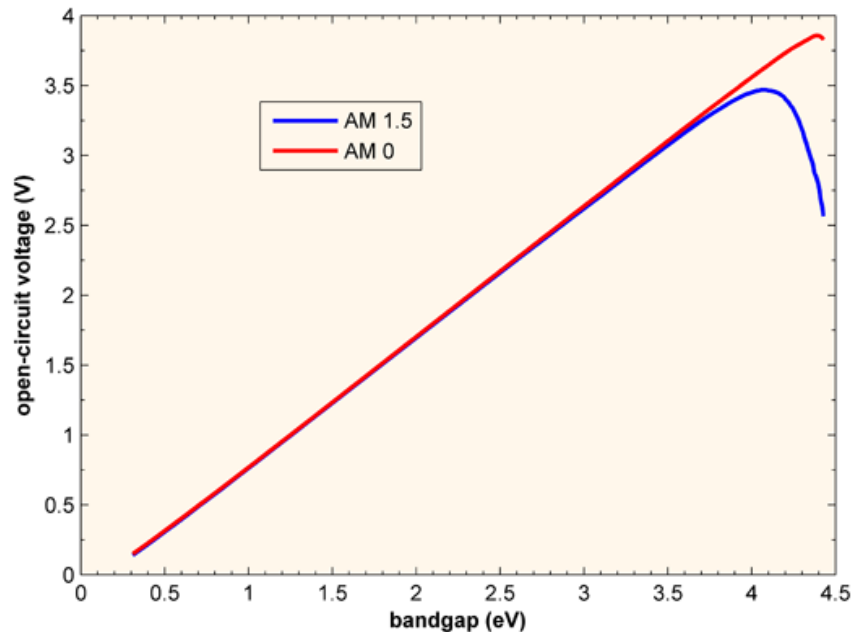
$$u = \frac{E_G}{kT}$$

- Evaluating the integral in the above equation is quite complex. The graph below uses the method outlined in the principle of detailed balance.



Diode saturation current as a function of band gap. The values are determined from detailed balance and place a limit on the open circuit voltage of a solar cell.

- The J_0 calculated above can be directly plugged into the standard solar cell equation (4.12) to determine the V_{OC} so long as the voltage is less than the band gap, as is the case under one sun illumination.

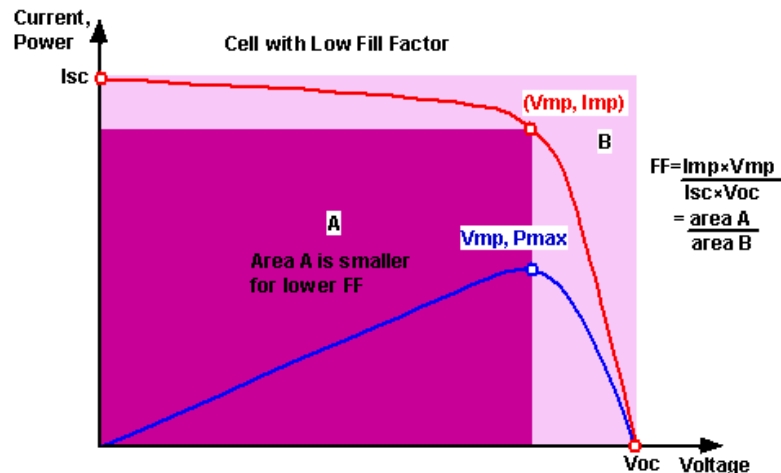


V_{OC} as function of bandgap for a cell with AM 0 and AM 1.5. The V_{OC} increases with bandgap as the recombination current falls. There is drop off in V_{OC} at very high band gaps due to the very low I_{SC} .

Fill Factor (FF)

- The J_{sc} and V_{oc} are the maximum current and voltage respectively from a solar cell. However, at both of these operating points, the power from the solar cell is zero.
- At particular current and voltage (I_{MPP} and V_{MPP}) amongst the data points in the fourth quadrant, the product of current and voltage gives the maximum power (P_{max}). This point is called maximum power point. The larger the area of rectangular ($I_{MPP} \times V_{MPP}$) the more the I-V curve resembles a rectangular with area $V_{oc} \times I_{sc}$. The ratio of these two areas is called fill factor (FF) which relates P_{max} with V_{oc} and I_{sc} . FF also represents the quality of the shape of I-V curve. Mathematically, FF can be written as:

$$FF = \frac{V_{MPP} I_{MPP}}{V_{oc} I_{sc}} \quad (4.15)$$



Power Conversion Efficiency

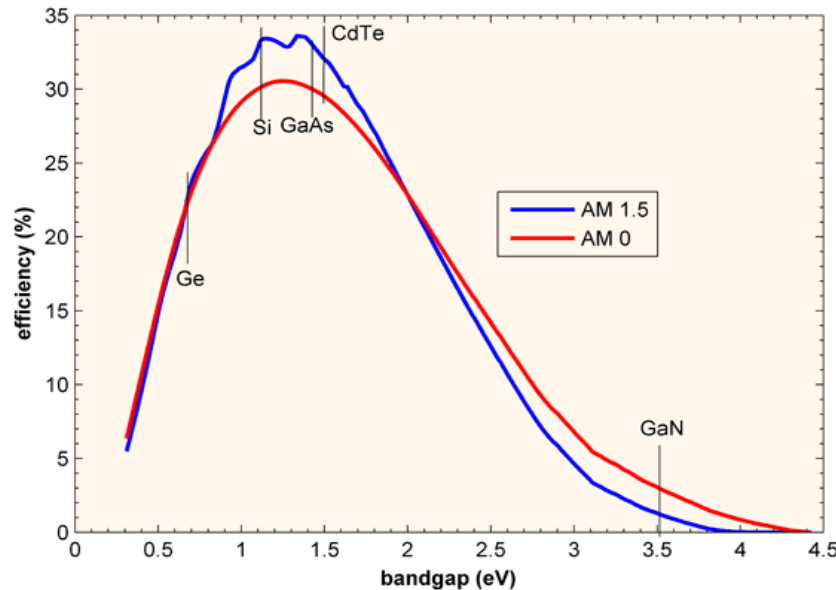
- The most important parameter that determines the performance of a solar cell is power conversion efficiency (PCE or η) which is defined as the ratio of the maximum output power to input power (P_{in}).

$$\eta = \frac{P_{\max}}{P_{in}} = \frac{V_{MPP} I_{MPP}}{P_{in}} = \frac{FF \times V_{oc} \times I_{sc}}{P_{in}} \quad (4.16)$$

- P_{in} is an important experimental parameter to compare the results of PCE and hence standard test condition should be used. This condition is set as AM1.5 spectrum illumination with an incident power density of 100mW/cm² at 25°C.
- Solar cells intended for space use are measured under AM0 conditions.

Limit of Efficiency of P-N Junction Solar Cells, Shockley-Queisser Limit

- As we discussed previously, the J_{sc} decreases with widening the bandgap whereas the V_{oc} increases with widening the bandgap. Because of the trade-off between these two parameters, there is an optimum bandgap for a single junction solar cell and the efficiency limit is predicted based on the optimum bandgap.
- William Shockley and Hans J. Queisser used the detailed balance technique to calculate the maximum efficiency of single junction solar cell. This maximum efficiency is well known as Shockley-Queisser Limit.



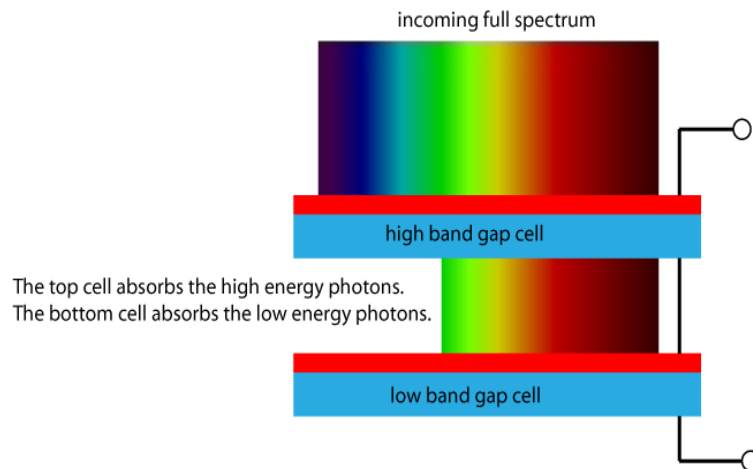
Limiting solar cell efficiency as a function of the material bandgap for one-sun illumination. The calculations assume that the only recombination is radiative. In actual devices the efficiencies are lower due to other recombination mechanisms and losses in parasitic resistances.

Further reading on efficiency limit

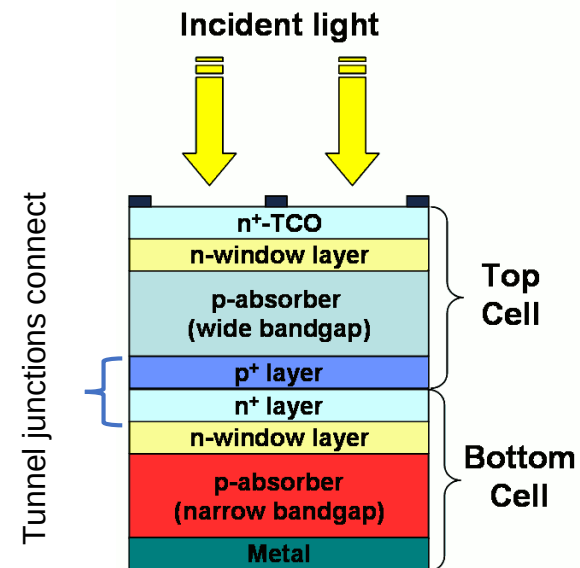
1. W. Shockley and Queisser, H. J., “**Detailed Balance Limit of Efficiency of p-n Junction Solar Cells**”, *Journal of Applied Physics*, vol. 32, pp. 510-519, 1961.
2. T. Tiedje, Yablonovich, E., Cody, G. D., and Brooks, B. G., “**Limiting Efficiency of Silicon Solar Cells**”, *IEEE TRANSACTIONS ON ELECTRON DEVICES*, vol. ED-31, 1984.

Tandem Cell or Multi-junction Cell

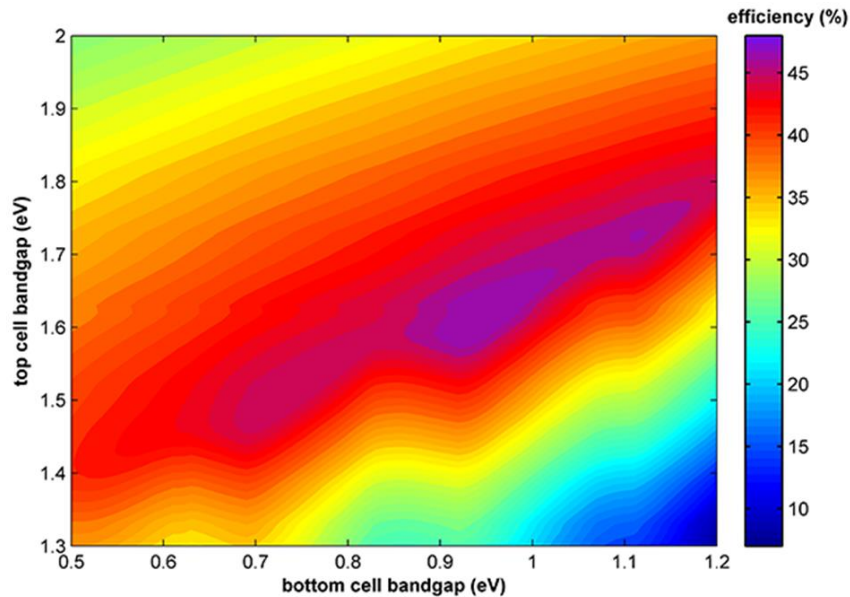
- One method to increase the efficiency of a solar cell is to split the spectrum and use a solar cell that is optimized to each section of the spectrum.
- Tandem solar cells can either be individual cells or connected in series. Series connected cells are simpler to fabricate but the current is the same through each cell so this constrains the band gaps that can be used. The most common arrangement for tandem cells is to grow them monolithically so that all the cells are grown as layers on the on substrate and tunnel junctions connect the individual cells.



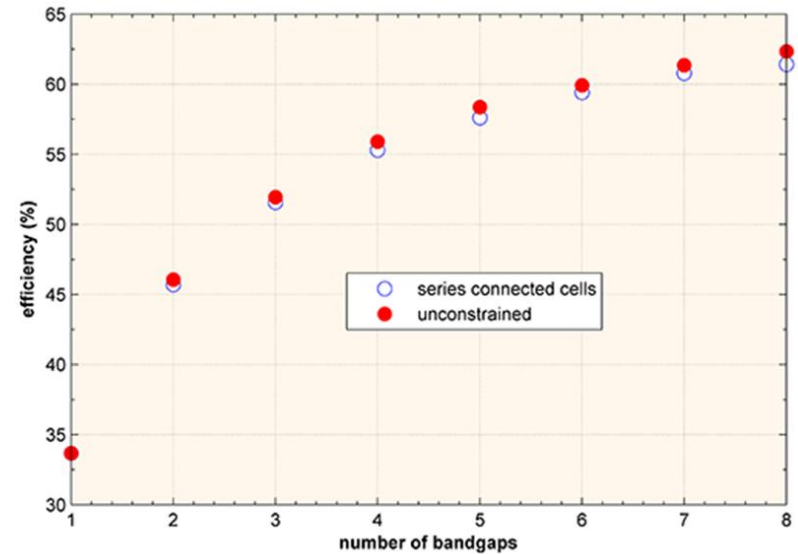
Series connected tandem solar cell. Adding more devices allows for each device to be optimized to a narrower spectrum giving a higher overall efficiency.



Monolithically grown tandem cell



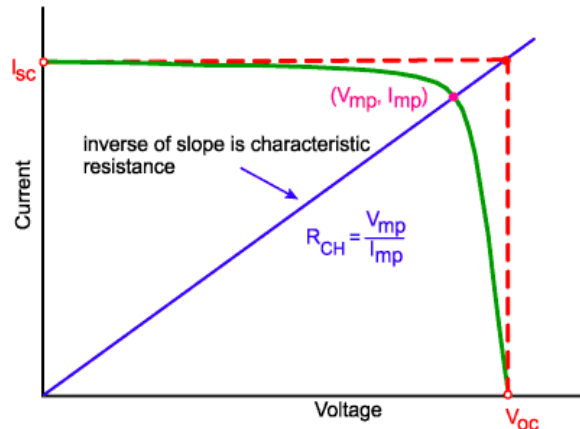
The maximum efficiency for a two junction tandem under the AM1.5G spectrum and without concentration is 47 %. At the peak efficiency the top cell has a bandgap of 1.63 eV and the bottom cell has a bandgap of 0.96 eV.



As the number of bandgaps increases the efficiency of the stack also potentially increases. Efficiency of a an ideal stack of solar cells as a function of the number of bandgaps

Characteristic Resistance

The characteristic resistance of a solar cell is the output resistance of the solar cell at its maximum power point. If the resistance of the load is equal to the characteristic resistance of the solar cell, then the maximum power is transferred to the load and the solar cell operates at its maximum power point. It is a useful parameter in solar cell analysis, particularly when examining the impact of parasitic loss mechanisms. The characteristic resistance is shown in the figure below.



$$R_{CH} = \frac{V_{MP}}{I_{MP}} \approx \frac{V_{OC}}{I_{SC}} \quad (4.17)$$

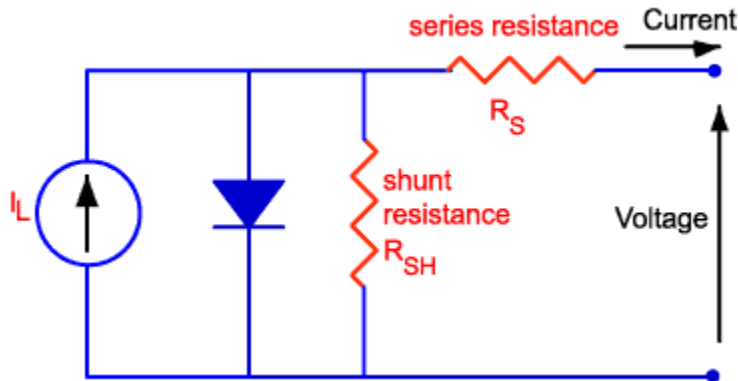
The characteristic resistance of a solar cell is the inverse of the slope of the line, shown in the figure above, which can be given as V_{MP} divided by I_{MP} for most cells it can be approximated by V_{OC} divided by I_{SC} .

Commercial silicon solar cells are typically very high current and low voltage. A 156 mm (6 inch) square solar cell has a current of almost 9 amps and a maximum power point voltage of 0.6 volts giving a characteristic resistance of around 0.067 Ω . As a consequence, connections to a single cells require resistances in the milliohm range.

Further Reading: M. A. Green, “**Accuracy of Analytical Expressions for Solar Cell Fill Factors**”, *Solar Cells*, vol. 7, pp. 337-340, 1982.

Effect of Parasitic Resistance

Resistive effects in solar cells reduce the efficiency of the solar cell by dissipating power in the resistances. The most common parasitic resistances are series resistance and shunt resistance. The inclusion of the series and shunt resistance on the solar cell model is shown in the figure below.



In most cases and for typical values of shunt and series resistance, the key impact of parasitic resistance is to reduce the fill factor. Both the magnitude and impact of series and shunt resistance depend on the geometry of the solar cell, at the operating point of the solar cell. Since the value of resistance will depend on the area of the solar cell, when comparing the series resistance of solar cells which may have different areas, a common unit for resistance is in Ωcm^2 . This area-normalized resistance results from replacing current with current density in Ohm's law as shown below:

$$R'(\Omega\text{cm}^2) = \frac{V}{J}$$

The effects of series and shunt resistances will be discussed in next section.

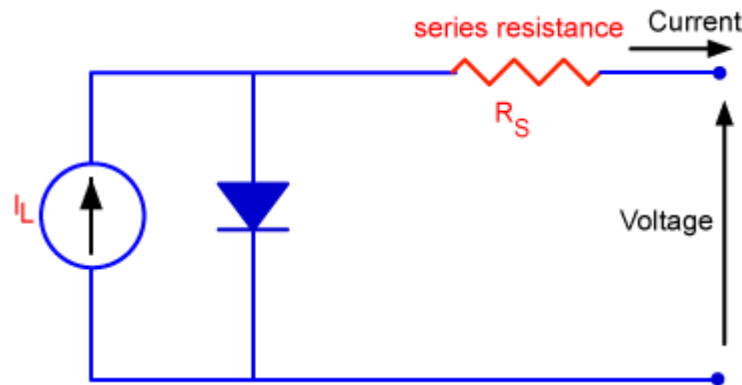
Series Resistance

Series resistance (R_s) is attributed to the finite conductivity of the semiconducting materials, the contact resistance between the semiconductors and the adjacent electrodes, and the resistance associated with electrodes and interconnections. R_s is also increased with a longer traveling distance of charge in e.g. thicker transport layers.

The main impact of series resistance is to reduce the fill factor, although excessively high values may also reduce the short-circuit current.

R_s can be estimated from the inverse slope at a positive voltage greater than V_{oc} where the I-V curves become linear.

$$R_s = \frac{dV}{dI} \quad V \geq V_{oc} \quad (4.18)$$

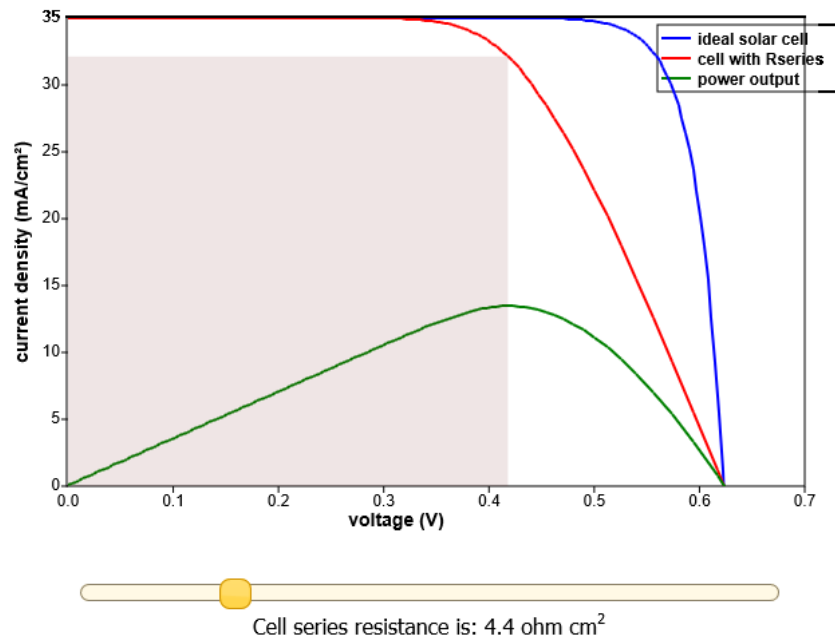


Schematic of a solar cell with series resistance.

- Incorporating R_s in solar cell, the current equation become

$$I = I_{ph} - I_0 \left[e^{\frac{q(V+IR_s)}{nkT}} \right]$$

- The effect of the series resistance on the IV curve is shown below.



- The main impact of R_s is to reduce the fill factor, although excessively high values may also reduce the short-circuit current.
- R_s also does not affect the open-circuit voltage of the solar cell.

An equation for the FF as a function of R_s can be determined for moderate values of series resistance. The maximum power may be approximated as the power in the absence of series resistance minus the power lost in the series resistance. The equation for the maximum power from a solar cell then becomes:

$$P'_{MP} \approx V_{MP} I_{MP} - I_{MP}^2 R_S = V_{MP} I_{MP} \left(1 - \frac{I_{MP}}{V_{MP}} R_S \right) = P_{MP} \left(1 - \frac{I_{SC}}{V_{OC}} R_S \right) \quad (4.19)$$

$$P'_{MP} = P_{MP} \left(1 - \frac{R_S}{R_{CH}} \right) \quad (4.20)$$

defining a normalized series resistance as;

$$r_S = \frac{R_S}{R_{CH}} \quad (4.21)$$

gives the following equation which approximates the effect of series resistance on the output power of a solar cell;

$$P'_{MP} = P_{MP} (1 - r_S) \quad (4.22)$$

Assuming that the V_{oc} and J_{sc} are not affected by the R_s allows the impact of series resistance on FF to be determined;

$$V'_{OC} I'_{SC} FF' = V_{OC} I_{SC} FF (1 - r_S) \quad (4.23)$$

$$FF' = FF (1 - r_S)$$

An empirical equation, which is slightly more accurate for the relationship between FF' and FF is;

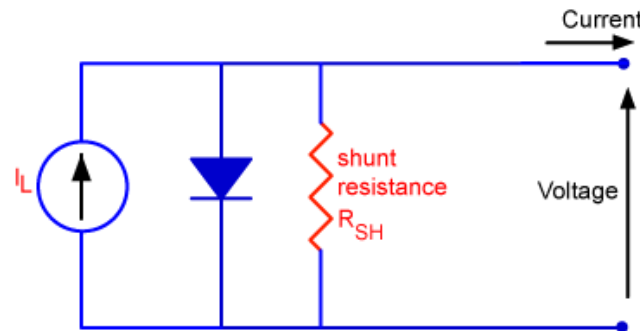
$$FF' = FF (1 - 1.1 r_S) + \frac{r_S^2}{5.4} \quad (4.24)$$

which is valid for $r_S < 0.4$ and $v_{oc} > 10$.

Shunt Resistance

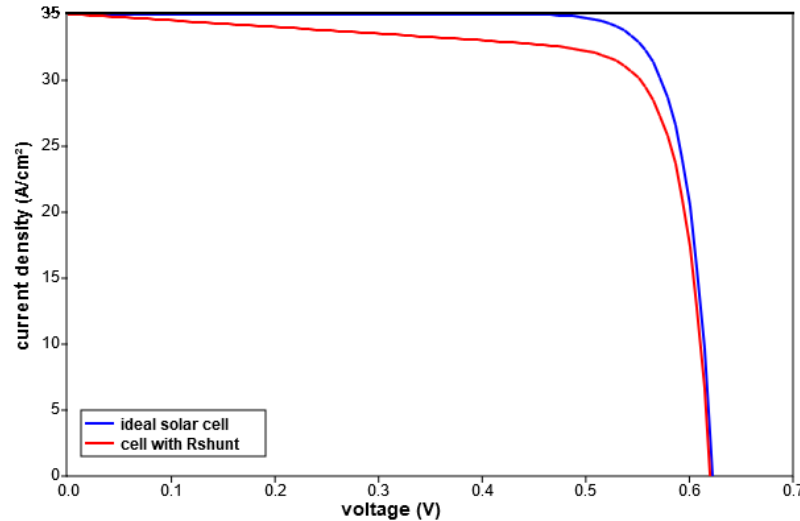
- Shunt resistance (R_{sh}) represents the loss of carriers via leakage paths; these may include pinholes in the film and the recombination and trapping of the carriers during their transit through the cell.
- Low R_{sh} causes power losses in solar cells by providing an alternate current path for the photo-generated current. Such a diversion reduces the amount of current flowing through the solar cell junction and reduces the voltage from the solar cell.
- The effect of a shunt resistance is particularly severe at low light levels, since there will be less photo-generated current. The loss of this current to the shunt therefore has a larger impact. In addition, at lower voltages where the effective resistance of the solar cell is high, the impact of a resistance in parallel is large.
- R_{sh} can be derived by taking the inverse slope of I-V curve around 0V:

$$R_{sh} = \frac{dV}{dI} \bigg|_{V=0} \quad (4.25)$$



The equation for a solar cell in presence of a shunt resistance is:

$$I = I_{ph} - I_0 \left[e^{\frac{qV}{nkT}} \right] - \frac{V}{R_{sh}} \quad (4.26)$$



Cell shunt resistance is: 200 ohm cm²

The impact of the shunt resistance on the fill factor can be calculated in a manner similar to that used to find the impact of series resistance on fill factor. The maximum power may be approximated as the power in the absence of shunt resistance, minus the power lost in the shunt resistance. The equation for the maximum power from a solar cell then becomes;

$$P'_{MP} \approx V_{MP} I_{MP} - \frac{V_{MP}^2}{R_{sh}} = V_{MP} I_{MP} \left(1 - \frac{V_{MP}}{I_{MP} R_{sh}} \right) = P_{MP} \left(1 - \frac{V_{OC}}{I_{SC} R_{sh}} \right) \quad (4.27)$$

$$P'_{MP} = P_{MP} \left(1 - \frac{R_{CH}}{R_{sh}} \right) \quad (4.28)$$

Defining a normalized shunt resistance as;

$$r_{SH} = \frac{R_{SH}}{R_{CH}} \quad (4.29)$$

Assuming that the open-circuit voltage and short-circuit current are not affected by the shunt resistance allows the impact of shunt resistance on FF to be determined as;

$$P'_{MP} = P_{MP} \left(1 - \frac{1}{r_{SH}} \right) \quad (4.30)$$

$$V'_{OC} I'_{SC} FF' = V_{OC} I_{SC} FF \left(1 - \frac{1}{r_{SH}} \right) \quad (4.31)$$

$$FF' = FF \left(1 - \frac{1}{r_{SH}} \right) \quad (4.32)$$

An empirical equation, which is slightly more accurate for the relationship between FF' and FF is;

$$FF' = FF \left(1 - \frac{V_{OC} + 0.7 FF}{V_{OC}} \frac{1}{r_{SH}} \right) \quad (4.33)$$

which is valid for $r_{sh} > 0.4$.

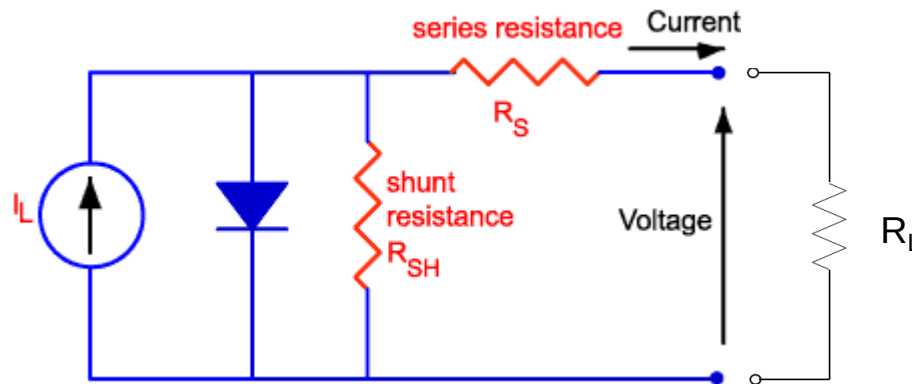
Typical values for area-normalized shunt resistance are in the $M\Omega\text{cm}^2$ range for laboratory type solar cells, and $1000 \Omega\text{cm}^2$ for commercial solar cells.

Impact of both series and shunt resistance

In the presence of both series and shunt resistances, the I-V curve of the solar cell is given by;

$$I = I_L - I_0 \exp \left[\frac{q(V + IR_S)}{nkT} \right] - \frac{V + IR_S}{R_{SH}} \quad (4.34)$$

and the circuit diagram of the solar cell is given as;



In principle, R_S needs to be minimized (ideally zero) so that most of the electrical potential is applied to the load. R_{SH} needs to be maximized (ideally infinity) so that all the current is passing through the load.

An external load resistor, R_L , represents that a solar cell is connected to an external load. Alternatively, a source meter can be connected to the cell when I-V characteristic of the cell is measured.

To combine the effect of both series and shunt resistances, the expression for FF_{sh} , derived above, can be used, with FF_0 replaced by FF_S . The overall equation then becomes;

$$FF_{SH} = FF_S \left(1 - \frac{V_{OC} + 0.7 FF_S}{V_{OC}} \frac{r_{SH}}{r_{SH}} \right) \quad (4.35)$$

where FF_S is given by

$$FF_S = FF_0 (1 - 1.1 r_S) + \frac{r_S^2}{5.4} \quad (4.36)$$

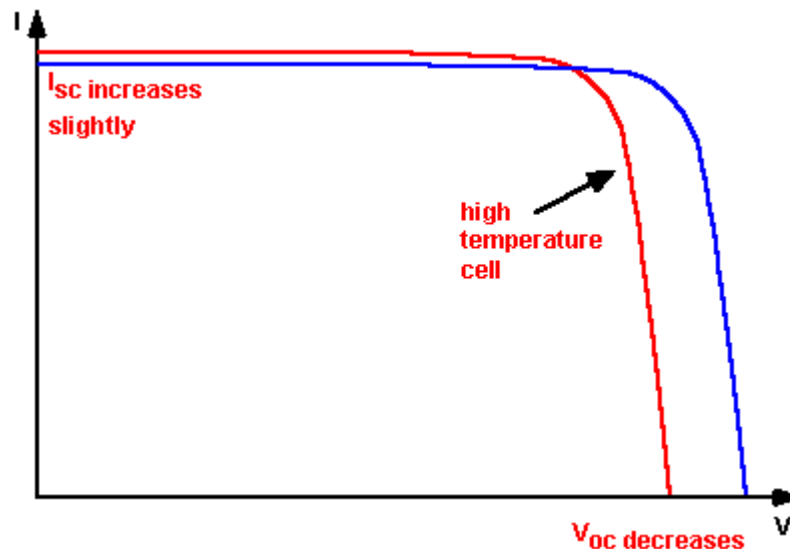
and by combining the above equations, the net equation for FF becomes;

$$FF = FF_0 \left\{ (1 - 1.1 r_S) + \frac{r_S^2}{5.4} \right\} \left\{ 1 - \frac{V_{OC} + 0.7 FF_0}{V_{OC}} \frac{r_{SH}}{r_{SH}} \left[(1 - 1.1 r_S) + \frac{r_S^2}{5.4} \right] \right\} \quad (4.37)$$

Effect of Temperature

Like all other semiconductor devices, solar cells are sensitive to temperature. Increases in temperature reduce the band gap of a semiconductor, thereby effecting most of the semiconductor material parameters. The decrease in the band gap of a semiconductor with increasing temperature can be viewed as increasing the energy of the electrons in the material. Lower energy is therefore needed to break the bond. In the bond model of a semiconductor band gap, reduction in the bond energy also reduces the band gap. Therefore increasing the temperature reduces the band gap.

In a solar cell, the parameter most affected by an increase in temperature is the open-circuit voltage. The impact of increasing temperature is shown in the figure below.



Effect of light intensity

Changing the light intensity incident on a solar cell changes all solar cell parameters, including the J_{sc} , V_{oc} , FF, the efficiency and the impact of series and shunt resistances. The light intensity on a solar cell is called the number of suns, where 1 sun corresponds to standard illumination at AM1.5, or 1 kW/m^2 . For example a system with 10 kW/m^2 incident on the solar cell would be operating at 10 suns, or at 10X. A PV module designed to operate under 1 sun conditions is called a "flat plate" module while those using concentrated sunlight are called "concentrators".

Concentrators

A concentrator is a solar cell designed to operate under illumination greater than 1 sun. The incident sunlight is focused or guided by optical elements such that a high intensity light beam shines on a small solar cell. Concentrators have several potential advantages, including a higher efficiency potential than a one-sun solar cell and the possibility of lower cost. The short-circuit current from a solar cell depends linearly on light intensity, such that a device operating under 10 suns would have 10 times the short-circuit current as the same device under one sun operation. However, this effect does not provide an efficiency increase, since the incident power also increases linearly with concentration. Instead, the efficiency benefits arise from the logarithmic dependence of the open-circuit voltage on short circuit. Therefore, under concentration, V_{oc} increases logarithmically with light intensity, as shown in the equation below;

$$V'_{oc} = \frac{nkT}{q} \ln\left(\frac{XI_{sc}}{I_0}\right) = \frac{nkT}{q} \left[\ln\left(\frac{I_{sc}}{I_0}\right) + \ln X \right] = V_{oc} + \frac{nkT}{q} \ln X \quad (4.38)$$

where X is the concentration of sunlight.

From the equation above, a doubling of the light intensity ($X=2$) causes a 18 mV rise in V_{OC} .

The cost of a concentrating PV system may be lower than a corresponding flat-plate PV system since only a small area of solar cells is needed.

The efficiency benefits of concentration may be reduced by increased losses in series resistance as the short-circuit current increases and also by the increased temperature operation of the solar cell. As losses due to short-circuit current depend on the square of the current, power loss due to series resistance increases as the square of the concentration.

Low Light Intensity

Solar cells experience daily variations in light intensity, with the incident power from the sun varying between 0 and 1 kW/m². At low light levels, the effect of the shunt resistance becomes increasingly important. As the light intensity decreases, the bias point and current through the solar cell also decreases, and the equivalent resistance of the solar cell may begin to approach the shunt resistance. When these two resistances are similar, the fraction of the total current flowing through the shunt resistance increases, thereby increasing the fractional power loss due to shunt resistance. Consequently, under cloudy conditions, a solar cell with a high shunt resistance retains a greater fraction of its original power than a solar cell with a low shunt resistance.

Ideality Factor

The ideality factor of a diode is a measure of how closely the diode follows the ideal diode equation. The derivation of the simple diode equation uses certain assumption about the cell. In practice, there are second order effects so that the diode does not follow the simple diode equation and the ideality factor provides a way of describing them.

Recombination mechanisms

The ideal diode equation assumes that all the recombination occurs via band to band or recombination via traps in the bulk areas from the device (i.e. not in the junction). Using that assumption the derivation produces the ideal diode equation below and the ideality factor, n , is equal to one.

$$I = I_L - I_0 \left[\exp \left(\frac{qV}{nkT} \right) - 1 \right]$$

However recombination does occur in other ways and in other areas of the device. These recombinations produce ideality factors that deviate from the ideal. Deriving the ideal diode equation by considering the number of carriers the need to come together during the process produces the results in the table below.

Recombination Type	Ideality factor	Description
SRH, band to band (low level injection)	1	Recombination limited by minority carrier.
SRH, band to band (high level injection)	2	Recombination limited by both carrier types.
Auger	2/3	Two majority and one minority carriers required for recombination.
Depletion region (junction)	2	two carriers limit recombination.

Take home message

- Charge generation
- Charge collection
- Quantum efficiency
- I-V characteristic
- Solar cell parameters
- Effect of parasitic resistance, temperature, light intensity
- Ideality factor